



MicroRNA-100-5p in exosomes from human placental mesenchymal stem cells: a key modulator of anti-psoriatic effects through mTOR/S6 K1 pathway

Yu-Chen Huang^{1,2,3} · Chao-Yuan Chang⁴ · Chun-Jen Huang^{1,5,6,7} 

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Abstract

Exosomes from mesenchymal stem cells (MSCs) demonstrate therapeutic potential against psoriasis, primarily due to their anti-inflammatory properties. However, the specific components within the exosomal cargo, particularly microRNAs, that mediate this activity remain unclear. This study investigated the role of a specific microRNA in mediating the therapeutic effects of exosomes from human placenta MSCs (MSC-exosomes) in an imiquimod (IMQ)-induced psoriasis murine model, with a focus on the underlying mechanisms, particularly the downstream pathways of the specific microRNA. Our small RNA sequencing analysis identified microRNA (miR)-100-5p as a potential target. Adult male Balb/c mice were then randomly assigned to the IMQ, IMQ plus MSC-exosomes (IMQ + Exo), IMQ plus MSC-exosomes treated with a specific miR-100-5p inhibitor (IMQ + Exoi), or IMQ plus agomiR-100-5p group (IMQ + AgomiR) (n = 6 per group). Control groups were also included. Mice were sacrificed, and psoriasis severity was assessed the day after 6 consecutive days of daily IMQ exposure, followed by 7 consecutive days of daily treatment. The IMQ + Exo group exhibited significantly lower epidermal thickness, levels of inflammatory cytokines and psoriasis-related cytokines (interleukin [IL]-12, IL-17, and IL-23), and expression of mammalian target of rapamycin (mTOR)/S6 kinase 1 (S6K1) (the predicted downstream target of miR-100-5p), compared to the IMQ group (all $p < 0.05$). Notably, inhibition of miR-100-5p nullified these therapeutic effects of MSC-exosomes. This study provides evidence that miR-100-5p plays a crucial role in mediating the therapeutic effects of MSC-exosomes in an IMQ-induced murine psoriasis model, potentially by modulating the mTOR/S6K1 pathway.

Key messages

- Exosomes from human placental mesenchymal stem cells demonstrate therapeutic potential against psoriasis.
- MicroRNA-100-5p plays a crucial role in mediating the therapeutic effects of mesenchymal stem cells exosomes in an imiquimod-induced murine psoriasis model, potentially by modulating the mammalian target of rapamycin/S6 kinase 1 pathway.

Keywords Exosome · Mesenchymal stem cell · MicroRNA-100-5p · MTOR · Psoriasis

✉ Chao-Yuan Chang
110234@w.tmu.edu.tw

✉ Chun-Jen Huang
cjhuang@tmu.edu.tw

¹ Graduate Institute of Clinical Medicine, College of Medicine, Taipei Medical University, 250 Wuxing Street, Xinyi District, Taipei 110, Taiwan

² Department of Dermatology, School of Medicine, College of Medicine, Taipei Medical University, Taipei, Taiwan

³ Department of Dermatology, Wan Fang Hospital, Taipei Medical University, Taipei, Taiwan

⁴ Department of Medical Research, Wan Fang Hospital, Taipei Medical University, Taipei, Taiwan

⁵ Research Center of Cell Therapy and Regeneration Medicine, Taipei Medical University, Taipei, Taiwan

⁶ Department of Anesthesiology, School of Medicine, College of Medicine, Taipei Medical University, Taipei, Taiwan

⁷ Department of Anesthesiology, Wan Fang Hospital, Taipei Medical University, Taipei, Taiwan

Introduction

Psoriasis is a chronic disease affecting the skin and joints, driven by genetic factors and inflammatory dysregulation [1]. It involves abnormal keratinocyte activity and immune cell infiltration, with T helper (Th) 1 cytokines like tumor necrosis factor (TNF)- α and interferon (IFN)- γ abundant in psoriatic plaques [1]. Key cytokines, including interleukin (IL)-12, IL-23, and IL-17 A, are found in the serum, skin lesions, and blood of psoriasis patients, contributing to its pathogenesis.¹

Exosomes, small extracellular vesicles (30–150 nm), facilitate intercellular communication by transferring substances like proteins, RNAs and microRNAs between cells [2]. They reflect the condition of their source cells and are implicated in the therapeutic effects of mesenchymal stem cells (MSCs), particularly in immunomodulation [3]. Studies have explored MSC-derived exosomes for treating skin inflammatory diseases, such as psoriasis [4, 5]. For example, in murine models, exosomes from human embryonic or umbilical cord-derived MSCs improved imiquimod (IMQ)-induced psoriasis-like dermatitis by reducing keratinocyte proliferation [5]. Intradermal injection of exosomes from human umbilical cord-derived MSCs suppressed the proliferation of keratinocytes in an IMQ-induced moderate psoriatic mouse model [6, 7]. However, the exact components responsible for these effects remain unknown.

MicroRNAs, conserved noncoding RNAs (19–25 nucleotides), regulate gene expression and are thought to mediate exosome effects [8, 9]. Over 250 microRNAs are differentially expressed in the skin and blood of psoriasis patients [10], making them potential therapeutic targets for exosome-based treatments. To investigate this, we hypothesized that a specific microRNA from human placenta MSC exosomes plays a key role in treating IMQ-induced psoriasis-like dermatitis in mice. Using small RNA sequencing, we identified this microRNA and tested its effects with an inhibitor and agomiR, focusing on its role in inflammation and downstream pathways.

Materials and methods

Isolation and purification of exosomes

Isolation and purification of exosomes from human placenta MSCs (generously provided by Professor Ye-Hua Huang, Taipei Medical University, Taiwan) was conducted through culture medium harvesting, centrifugation, supernatant collection, filtration, and

ultracentrifugation (Beckman Coulter Optima L-90 K Ultracentrifuge; 100,000 $\times$ g; 4 $^{\circ}$ C; 90 min; Type 50.2 Ti rotor, k-factor: 157.7; Beckman Coulter Inc., Brea, CA, USA), as per previous protocols [11, 12]. The resulting pellets were resuspended, pooled, ultracentrifuged, resuspended again, purified, and ultracentrifuged again. The top fractions of the gradient were collected, diluted, and centrifuged. The pellets were resuspended and stored at -80° C.

Sizing and morphology analyses of MSC-exosomes

The size distribution of MSC-exosomes was detected by nanoparticle tracking analysis (NTA, ZetaView PMX 110, Particle Metrix), as per the manufacturer's protocols. In addition, an MSC-exosome suspension was fixed, transferred to grids (Polysciences, Warrington, PA, USA), dried, and the morphology was examined by transmission electron microscopy (TEM) (Hitachi HT-7700; Hitachi, Tokyo, Japan) [13].

Marker analysis of MSC-exosomes using immunoblotting assay

Markers of MSC-exosomes, including CD63 and ALIX [14], were detected by immunoblotting assay. In brief, proteins from MSC-exosomes were extracted, segregated through electrophoresis, and transferred onto nitrocellulose membranes (Bio-Rad Laboratories, Hercules, CA, USA) [15]. Membranes were incubated with primary antibodies against CD63 (anti-CD63 antibody; 25682-1-AP; Proteintech, Rosemont, IL, USA) and ALIX (anti-ALIX antibody, ab235377; Abcam, Cambridge; UK), respectively. Bound antibodies were detected through chemiluminescence (ECL Plus kit; Amersham Bioscience, Buckinghamshire, UK).

Biodistribution of MSC-exosomes

To confirm the biodistribution of MSC-exosomes, MSC-exosomes (1×10^8 particles per mouse) were conjugated with Cy7 mono NHS ester (Amersham) and topically applied to the shaved skin on the back of mice. Three mice were euthanized by decapitation at each time point (0, 2, 24, and 48 h) following MSC-exosome administration. Their tissues were collected and analyzed using an ex vivo bioluminescence imaging assay (IVIS Lumina XRMS and Living Image software; PerkinElmer, Waltham, MA, USA) [15]. The MSC-exosome dose was based on previous data showing its ability to mitigate endotoxin-induced organ injuries [16].

Additionally, immunofluorescence assay was employed to detect CD63 [14], to confirm MSC-exosome distribution in the skin tissues after 6 days of IMQ application and followed by 7 days of MSC-exosome or PBS treatment. The

formaldehyde-fixed samples were embedded in paraffin wax (Sigma-Aldrich, Burlington, MA, USA), deparaffinized, rehydrated, and incubated with a primary antibody against CD63 (anti-CD63 antibody, 25,682–1-AP; Proteintech) and observed using a confocal microscope (LSM-780; Zeiss, Oberkochen, Germany).

Animals care and the IMQ-induced psoriasis murine model

Eight-week-old male Balb/c mice (the National Laboratory Animal Center in Taipei, Taiwan) were housed with free access to food and water under a 12-h light/dark cycle. Animal care followed the U.S. National Institutes of Health (NIH) guidelines, and the study was approved by Taipei Medical University's Institutional Animal Care and Use Committee (approval: LAC-2021–0500). The mice were weighed, and 1.5 cm × 2 cm of the back skin were shaved for IMQ induction. A psoriatic phenotype was induced by daily topical applications of 50 mg IMQ cream (5% Aldara cream; Ensign Laboratories Pty Ltd., Mulgrave, VIC, Australia) on the shaved back for 6 consecutive days, as per a previous protocol [5].

Analysis of the microRNAs in MSC-exosomes and skin tissues of mice

For identification of the therapeutic microRNAs in the MSC-exosomes, microRNAs in MSC-exosomes and skin tissues were analyzed via small RNA sequencing [17]. To assess microRNA expression in skin tissues, four mice received daily topical IMQ for six days (IMQ group), and three received phosphate-buffered saline (Sham group). Daily IMQ application for 6 days resulted in an approximate 12% mortality rate; therefore, four mice were included in the IMQ group to ensure three survivors for analysis. Surviving mice ($n = 3$ per group) were euthanized, and their skin tissues were collected for small RNA sequencing.

For sequencing, 100 ng of total RNA from MSC-exosomes and skin tissues were processed. Libraries were prepared using the QIAseq® microRNA Library Kit (QIAGEN Inc., Hilden, German). After adaptor ligation, cDNA synthesis, PCR amplification, and size selection (170–200 bp), libraries were assessed for quality and sequenced on the Illumina NovaSeq 6000 platform (Illumina Inc., San Diego, CA, USA), generating 75 bp single-end reads to identify microRNAs.

Identification of the potential microRNA(s)

Small RNA sequencing data were analyzed to identify differentially expressed microRNAs between the IMQ and Sham groups. A volcano plot showed up- and down-regulated

microRNAs, while heatmaps illustrated the most significantly down-regulated microRNAs in the IMQ group.

MicroRNAs from MSC-exosomes were compared with those differentially expressed in skin tissues. Overlapping microRNAs, particularly those downregulated in the IMQ group and present in MSC-exosomes, were identified as potential mediators of MSC-exosome therapeutic effects. Bioinformatic analyses using miRDB, Targetscan, and miR-TarBase identified key targets relevant to psoriasis.

Generating the inhibitor-treated MSC-exosomes with specific microRNA inhibition

Our analysis identified miR-100-5p as a potential mediator of MSC-exosome therapeutic effects. To investigate the role of miR-100-5p, we used a miR-100-5p inhibitor (5'-AACCCGUAGAUCCGAACUUGUG-3'; Integrated DNA Technologies, Inc., Coralville, IA, USA) to see whether the therapeutic effect of MSC-exosomes was reversed. Electroporation of the miR-100-5p inhibitor into exosomes was performed through precipitation, resuspension, transferal, electroporation (150 V/100 μ F), removal of free-floating microRNAs, and ultracentrifugation. The processed pellet was stored at -80°C .

AgomiR-100-5p synthesis

The role of miR-100-5p was also investigated by employing an agomiR-100-5p for treating IMQ-induced psoriasiform dermatitis. The agomiR-100-5p was synthesized (Integrated DNA Technologies) with the target sequences for agomiR-100-5p, sense: 5' AACCCGUAGAUCCGAA CUU GUG –3', and antisense: 5'-CACAAAGUUCGGAUCUACG GGUU-3'.

Study design

In order to confirm the role of miR-100-5p in MSC-derived exosome therapy for psoriasis, we randomly divided the mice into six groups: Sham, Exo, IMQ, IMQ + Exo, IMQ + Exoi, and IMQ + AgomiR. Sham and Exo groups received topical PBS for 6 days, followed by PBS or MSC-derived exosomes (1×10^8 particles per mouse) for 7 days. IMQ, IMQ + Exo, and IMQ + Exoi groups received topical IMQ for 6 days, followed by PBS, MSC-derived exosomes (1×10^8 particles per mouse), or inhibitor-treated MSC-exosomes (1×10^8 particles per mouse) for 7 days. The IMQ + AgomiR group received IMQ for 6 days, followed by intradermal agomiR-100-5p injections (2 nmol per mouse). Mortality rates were 0% for the Sham and Exo groups, and 33.3%, 25%, 40%, and 33.3% for the IMQ, IMQ + Exo, IMQ + Exoi, and IMQ + Agomir groups, respectively. To ensure 6 survivors per group, the initial group sizes were 6, 6, 9, 8,

10, and 9. On the day after the final treatment, the psoriasis severity scores, the spleen and body weight, the histological analysis, and cytokines of skin tissues were evaluated. The details were described in the following paragraphs.

Psoriasis area and severity index (PASI) score of animal

PASI score of animal, including scoring of erythema, scaling, and thickening (0–4 scale), was used to assess IMQ-induced psoriatic dermatitis severity, as reported previously [5, 6]. The cumulative score (0–12) was calculated to measure overall severity.

Euthanizing and tissue harvesting

One day after the final treatment, all surviving mice were weighed and euthanized by decapitation. Skin tissues from the back were harvested and then then dissected, with one half snap-frozen and stored at -80°C , and the other half placed in 37% formaldehyde solution (Sigma-Aldrich). The spleen was also harvested, and the spleen/body weight ratio was calculated.

Histological analysis

The formaldehyde-fixed skin tissue samples were embedded in paraffin wax, serially sectioned, and then stained with hematoxylin and eosin solution (Sigma-Aldrich). Epidermal thickening was observed under a light microscope (Moti-easyScan Pro 6; Motic Asia, Hong Kong). The thickness of the epidermis was measured using ImageJ software (Version 1.52e; free software by NIH, USA; available at <https://imagej.nih.gov/ij/>; accessed date: 1 August 2021).

Immunohistochemistry staining

Immunohistochemistry was used to assess Ki-67, a marker for cell proliferation [18], in skin samples. Bioinformatic analysis predicted mTOR as a target of miR-100-5p, so phosphorylated mTOR (p-mTOR) and phosphorylated S6 kinase 1 (p-S6 K1, a downstream mTOR component) [19] were also examined. Formaldehyde-fixed, paraffin-embedded tissues were prepared and stained with antibodies against Ki-67 (anti-Ki-67 antibody, IRQ 6001; iReal Technology, Hsinchu, Taiwan), phosphorylated mTOR (anti-p-mTOR antibody, GTX132803; GeneTex, Inc, California, USA), and phosphorylated S6 K1 (anti-p-S6 K1 antibody, GTX635621; GeneTex). Microscopy was conducted at 200 $\times$ magnification and signal densities were quantified using ImageJ software from six fields per section.

Enzyme-linked immunosorbent assay (ELISA)

Freshly frozen skin tissues were homogenized and centrifuged, and the supernatants were collected [15]. The concentrations of cytokines of skin tissues including in TNF- α (DY410-05), IFN- γ (DY485), IL-12 (DY419), IL-17 A (DY421), and IL-23 (DY1887) were measured using ELISA kits (all from R&D Systems, Emeryville, CA, USA).

Statistical analysis

A one-way analysis of variance was conducted to assess group differences. Data were presented as means with standard deviations. A significance level of $p < 0.05$ was used. The analysis was carried out using GraphPad Prism 10 (version 10.1.0 (316), 2023) for Windows.

Results

Confirmation and biodistribution of MSC-exosomes

MSC-exosomes were confirmed by their characteristic double-layer, cup-shaped morphology (Fig. 1a), 100–150 nm particle size (Fig. 1b), and positive detection of Alix and CD63 markers (Fig. 1c). Bioluminescence imaging revealed significantly higher Cy7-conjugated MSC-exosome signals in the skin at 2 and 24 h post-application compared to 0 h (both $p < 0.001$; Fig. 1d). Signals in other organs remained low and unchanged over time.

Immunofluorescence staining on the day after finished 6-day IMQ and 7-day MSC-exosome or PBS treatment confirmed MSC-exosome distribution throughout the epidermis in IMQ + Exo mice, and absent in IMQ-alone mice (Fig. 1e).

Identification of the target microRNA in the MSC exosomes

Small RNA sequencing identified 981 microRNAs with significantly different expression ($p < 0.05$) between the IMQ and Sham groups. A volcano plot (Fig. 2a) shows upregulated and downregulated microRNAs, while a heatmap reveals 16 significantly downregulated microRNAs (with expression $< 50\%$ of the Sham group) in IMQ-treated mice, with high expression ($> 10,000$ counts) in skin tissues.

Six downregulated microRNAs in IMQ-treated skin overlapped with those detected in MSC-exosomes. The amount of miR-100-5p is most abundant in the exosomes among the six microRNAs (Fig. 2b). Additionally, bioinformatic analysis predicted mTOR as a downstream target of miR-100-5p (Fig. 2c) and the role of mTOR in psoriasis was well established [20]. All of above suggested miR-100-5p might play a role in the therapeutic effects of MSC-exosomes.

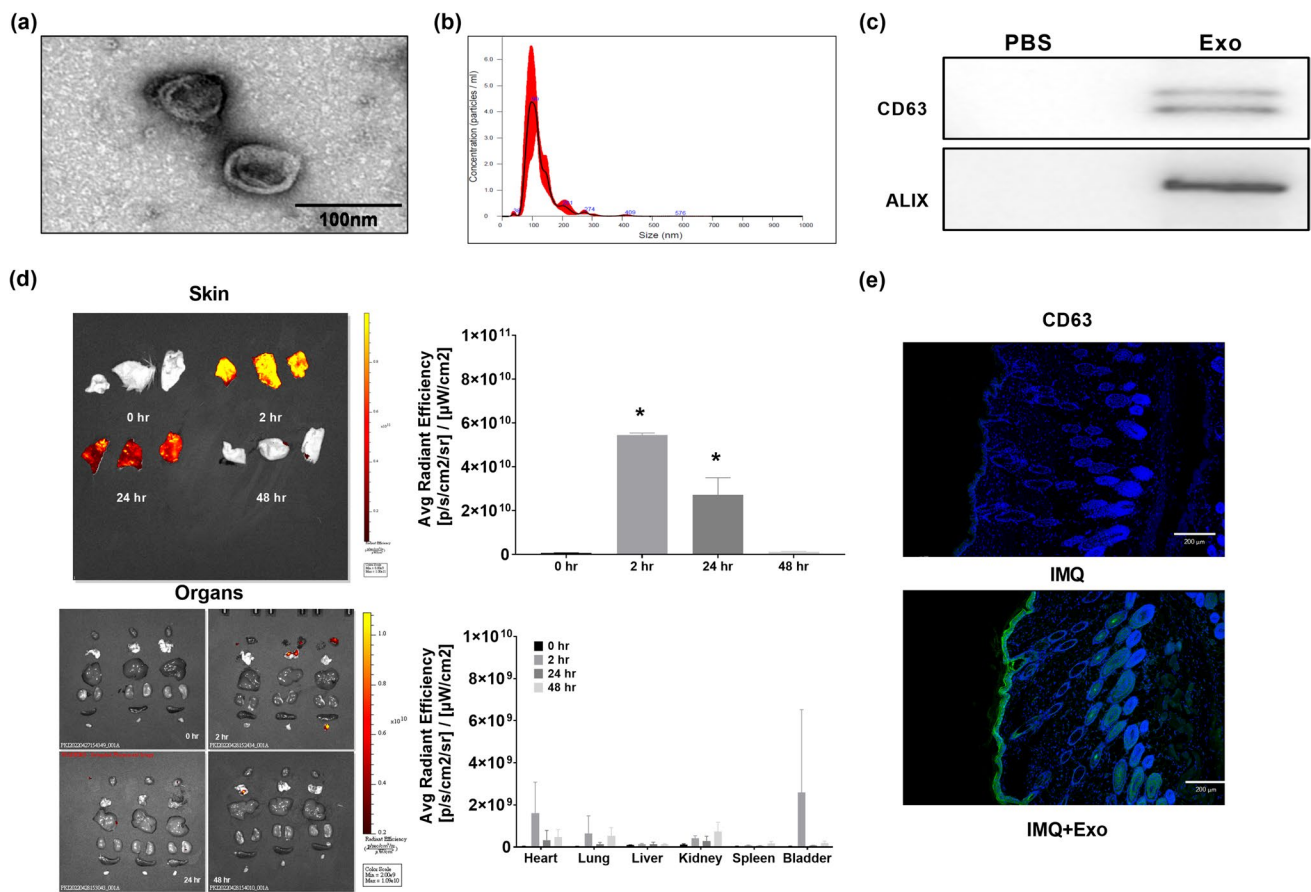


Fig. 1 Confirmation and biodistribution of exosomes from human placenta mesenchymal stem cells (MSC-exosomes). (a) Representative transmission electron microscopic images (60,000 ×) of MSC-exosomes. (b) Particle and sizing analysis of MSC-exosomes. (c) Representative gel photograph of MSC-exosome markers CD63 and Alix, detected using an immunoblotting assay. (d) Biodistribution of MSC-exosomes (conjugated with Cy7 mono NHS ester, 1×10^8 particles per mouse) in mice measured before (0 h) and at 2, 24, and 48 h after topical MSC-exosome administration, determined using an

ex vivo bioluminescence imaging assay. Biodistribution data were derived from 3 mice at each time point and presented as the mean $\pm$ standard deviation. * $p < 0.05$, versus the signal intensity measured at 0 h. (e) Representative immunofluorescence photography revealed the surface marker CD63 of MSC-exosomes, distributed over whole layer of epidermis in the mice receiving imiquimod (IMQ) plus MSC-exosomes (IMQ + Exo) but showed no signal in the IMQ alone mice. PBS: phosphate-buffered saline. Exo: MSC-exosomes

miR-100-5p mediates effects of MSC-exosomes on mitigating IMQ-induced psoriasis-like dermatitis in mice

Supplementary Table S1 presents PASI scores of animal before, during, and after interventions. The Sham and Exo groups showed no psoriasis-like symptoms, maintaining scores of 0.00 ± 0.00 throughout. The IMQ group had significantly elevated scores on Day 1 (7.83 ± 1.07 , $p < 0.001$), which decreased to 4.00 ± 0.00 by Day 4 and 0.00 ± 0.00 by Day 8. The IMQ + Exo group had similar Day 1 scores but showed a greater reduction by Day 4 (2.00 ± 0.00 , $p = 0.035$) (Supplementary Fig. 1a), indicating enhanced efficacy. The IMQ + Exoi group displayed a similar trend to the IMQ + Exo group, and the IMQ + AgomiR

group displayed a similar trend to the IMQ group. The spleen over body weight were also lower among IMQ + Exo group compared to the IMQ group. ($p < 0.0001$) (Supplementary Fig. 1b).

Figure 3a shows epidermis thickness data. Both the Sham and Exo groups had low thickness, while the IMQ group was significantly thicker than Sham ($p < 0.001$). The IMQ + Exo group showed reduced thickness compared to IMQ ($p < 0.001$), and the IMQ + Exoi group was thicker than IMQ + Exo ($p < 0.001$). The IMQ + AgomiR group had lower thickness than IMQ ($p = 0.003$) but was still thicker than IMQ + Exo ($p = 0.012$). Figure 3b displays Ki67 expression, which followed a similar trend as epidermal thickness, except there was no significant difference between IMQ + Exo and IMQ + Exoi ($p = 0.364$).

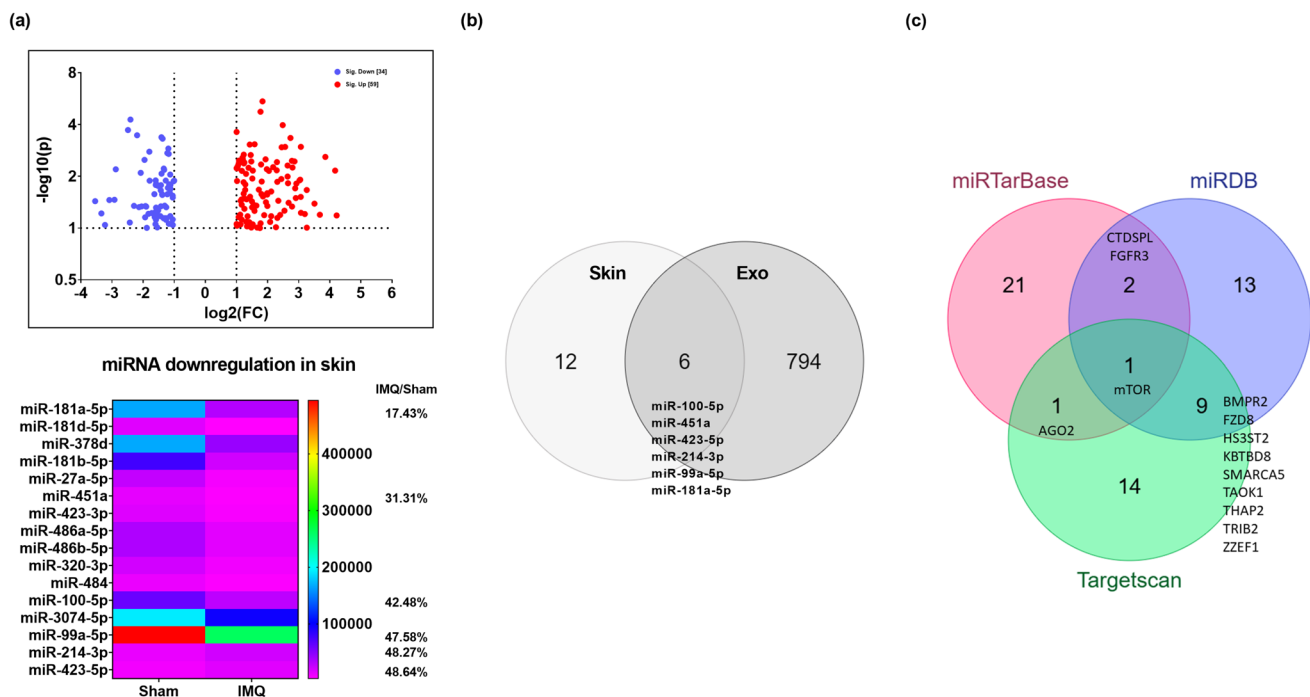


Fig. 2 Identification of target microRNAs (miRNAs) with therapeutic potential, achieved by examining miRNAs down-regulated in skin tissue samples from the IMQ group (compared to the Sham group) that overlap with those found in exosomes from human placenta mesenchymal stem cells (MSC-exosomes). (a) Up- and down-regulated miRNAs were analyzed via small RNA sequencing in skin tissue samples from the IMQ group ($n = 3$) and the Sham group ($n = 3$), presented in a volcano plot (upper). The heatmap (lower) showed 16 miRNAs that were significantly down-regulated in the IMQ group

(with expression $< 50\%$ of the Sham group) and with a high expression level ($> 10,000$ counts) in skin tissues. (b) A Venn diagram highlighted six miRNAs in MSC-exosomes that were overlapped with the down-regulated miRNAs in the IMQ group. The amount of miR-100-5p is most abundant in the exosomes among the six miRNAs (c) Another Venn diagram showed the predicted target miR-100-5p, identified by three bioinformatic tools. IMQ: the imiquimod group; Sham: the Sham group; Skin: skin tissues of the IMQ group; Exo: exosomes from human placenta mesenchymal stem cells

miR-100-5p mediates effects of MSC-exosomes on mitigating IMQ-induced cytokine upregulation in skin tissues of mice

Figure 4 presents TNF- α data, with the Sham and Exo groups having low levels. The IMQ group showed significantly higher TNF- α than Sham ($p < 0.001$). TNF- α levels were lower in IMQ + Exo compared to IMQ ($p < 0.001$), while IMQ + Exo had higher levels than IMQ + Exo ($p = 0.001$). IMQ + AgomiR also showed reduced TNF- α compared to IMQ ($p = 0.001$), with no significant difference between IMQ + Exo and IMQ + AgomiR ($p = 0.525$). Other cytokines (INF- γ , IL-12, IL-17 A, IL-23) followed similar trends, except for INF- γ showed no significant differences between IMQ and IMQ + AgomiR groups ($p = 0.066$) and IL-23 showed no significant differences between IMQ + Exo and IMQ + Exo groups. ($p = 0.079$).

miR-100-5p mediates effects of MSC-exosomes on mitigating IMQ-induced mTOR/S6 K1 upregulation in mice

Figure 5a shows p-mTOR levels. The Sham and Exo groups had low p-mTOR, while the IMQ group had significantly elevated levels ($p < 0.001$). IMQ + Exo had lower p-mTOR than IMQ ($p < 0.001$), but IMQ + Exo showed higher levels than IMQ + Exo ($p < 0.001$). IMQ + AgomiR reduced p-mTOR compared to IMQ ($p < 0.001$), with no significant difference between IMQ + Exo and IMQ + AgomiR ($p = 0.746$). A similar pattern was observed for p-S6 K1 (Fig. 5b), except the difference between IMQ + Exo and IMQ + Exo was not significant ($p = 0.054$).

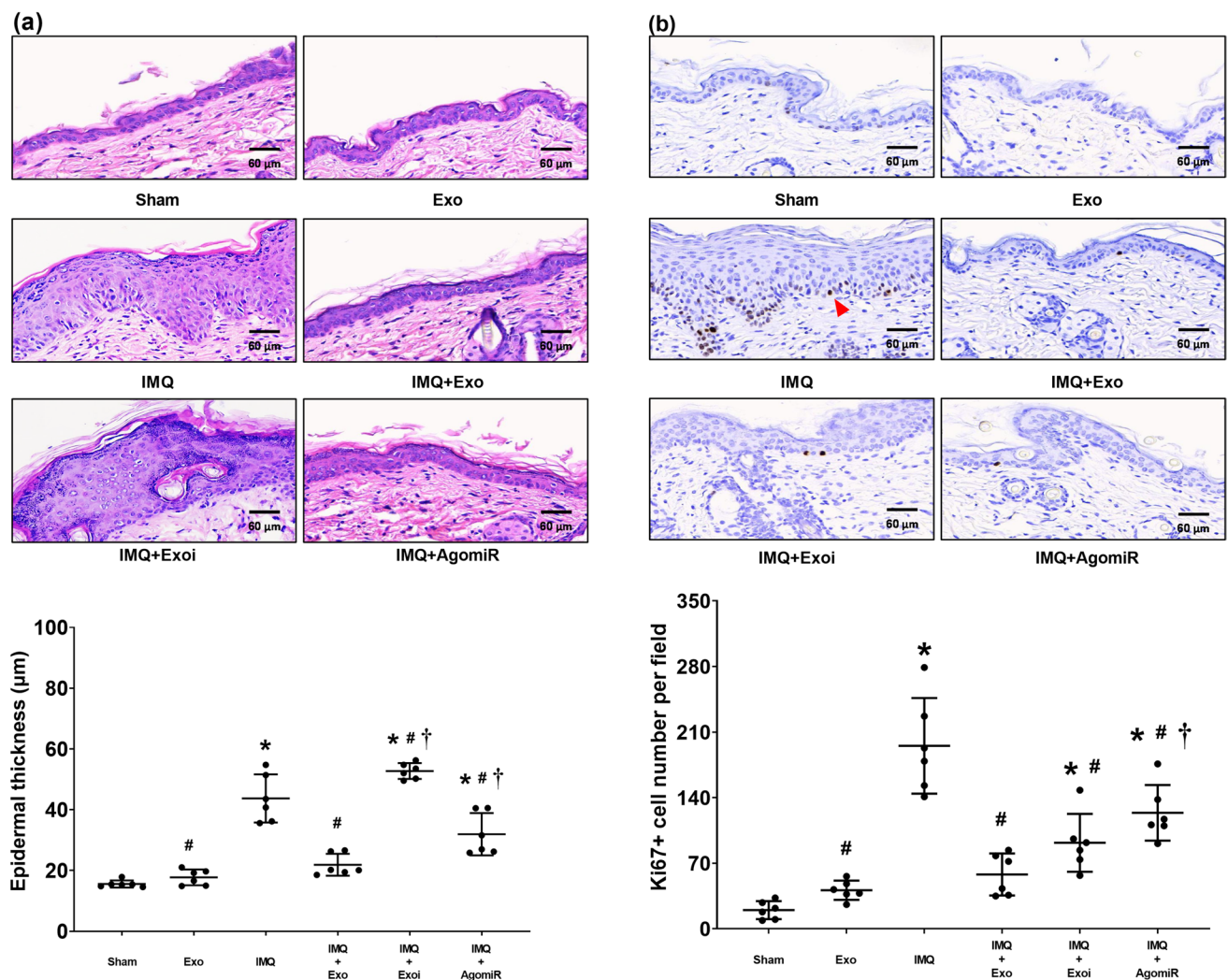


Fig. 3 The role of miR-100-5p in mediating the therapeutic effects of exosomes from human placenta mesenchymal stem cells (MSC-exosomes) in alleviating imiquimod (IMQ)-induced psoriasis-like dermatitis in mice. (a) Representative microscopic images of skin tissues stained with hematoxylin–eosin (200 × magnification) and measurements of epidermal thickness are shown. (b) Representative microscopic images of Ki67, a marker for cell proliferation (indicated by red arrows), along with the quantified total intensities of Ki67 in mouse skin tissues, assessed via immunohistochemistry, are pre-

sented. Tissue samples were collected the day after completing seven consecutive days of daily intervention. Data from six mice per group are presented as mean $\pm$ standard deviation. * $p < 0.05$ vs. Sham group; # $p < 0.05$ vs. IMQ group; † $p < 0.05$ vs. IMQ + Exo group. Sham: Sham group; Exo: MSC-exosome group; IMQ: imiquimod group; IMQ + Exo: IMQ plus MSC-exosomes group; IMQ + Exoi: IMQ plus MSC-exosomes treated with a miR-100-5p inhibitor; IMQ + AgomiR: IMQ plus agomiR-100-5p

Discussion

The data from this study provide clear evidence of the therapeutic effects of exosomes from human placenta MSCs (MSC-exosomes) against psoriasis. Specifically, our findings show that topical administration of MSC-exosomes improved epidermal hyperplasia and reduced inflammatory cytokines in mice with moderate psoriasis-like dermatitis induced by IMQ. Given the current lack of effective therapies for psoriasis, our findings have significant clinical implications and warrant further investigation.

The composition of MSC-exosomes is complex, and the specific components responsible for their biological activities are still being elucidated. Among the various cargo carried by exosomes, microRNAs have garnered the most attention due to their crucial role in regulating immune responses [21]. Psoriasis, on the other hand, is well known to be associated with the dysregulation of microRNAs, which can influence keratinocyte proliferation and differentiation processes [22]. Based on these findings, we hypothesized that MSC-exosomes may exert their therapeutic effects against psoriasis by delivering specific microRNAs, particularly

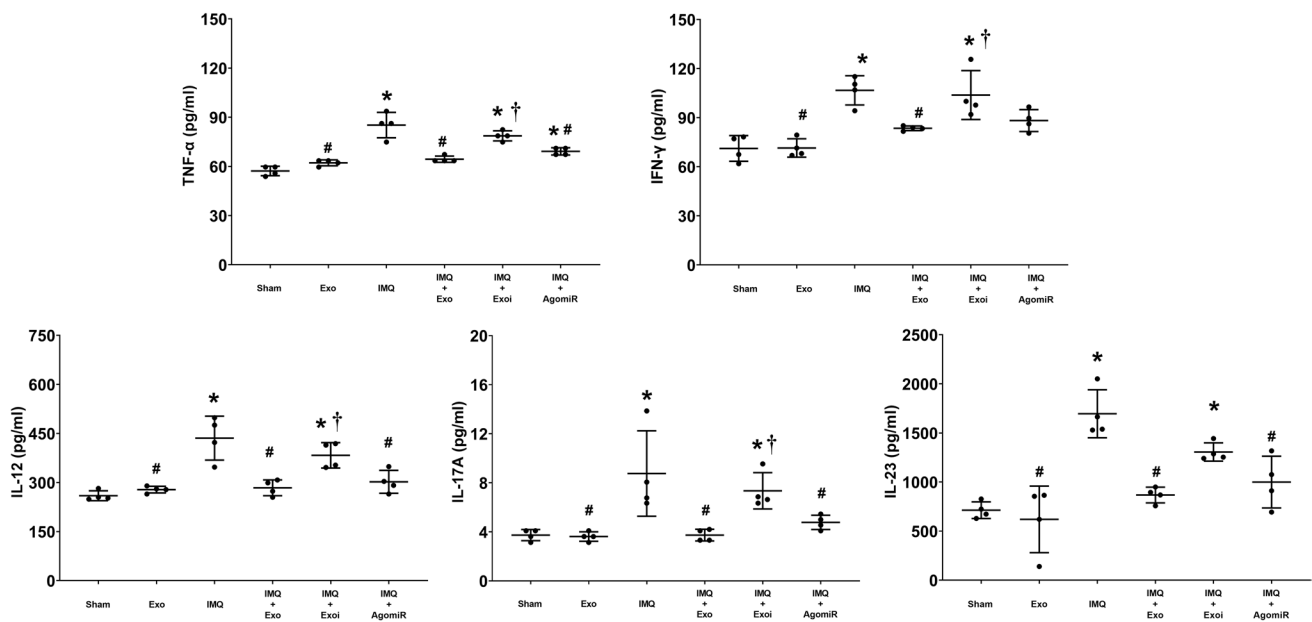


Fig. 4 The role of miR-100-5p in mediating the therapeutic effects of exosomes from human placenta mesenchymal stem cells (MSC-exosomes) in reducing imiquimod (IMQ)-induced cytokine upregulation in mouse skin tissues. Tissue samples were collected the day after completing seven consecutive days of daily intervention. Cytokine concentrations were measured using enzyme-linked immunosorbent assay. Data from four mice per group are presented as

mean ± standard deviation. TNF-α: tumor necrosis factor-α; INF-γ: interferon-γ; IL-12: interleukin-12; IL-17 A: interleukin-17 A; IL-23: interleukin-23. * $p < 0.05$ vs. Sham group; # $p < 0.05$ vs. IMQ group; † $p < 0.05$ vs. IMQ + Exo group. Sham: Sham group; Exo: MSC-exosome group; IMQ: imiquimod group; IMQ + Exo: IMQ plus MSC-exosomes group; IMQ + Exoi: IMQ plus MSC-exosomes treated with a miR-100-5p inhibitor; IMQ + AgomiR: IMQ plus agomiR-100-5p

through replenishing those that are downregulated. This concept is supported by our experimental data: inhibition of miR-100-5p significantly reduced the therapeutic effects of MSC-exosomes against IMQ-induced psoriasis-like dermatitis in mice, while intradermal injection of agomiR-100-5p replicated the therapeutic benefits of MSC-exosomes. Taken together, these results confirm our hypothesis and demonstrate that miR-100-5p plays a pivotal role in mediating the therapeutic effects of human placenta MSC-exosomes against IMQ-induced psoriasis-like dermatitis in mice.

One might wonder why MSC-exosomes from human placenta exhibit significant therapeutic effects against psoriasis, particularly when tested in murine models. A key factor could be the involvement of miR-100-5p, which appears to play a pivotal role in both species. In alignment with our findings, previous studies have demonstrated that miR-100-5p is downregulated in patients with psoriasis [23]. Formerly known as miR-100, miR-100-5p belongs to the miR-99 family, which includes miR-99a and miR-99b. This family is among the most evolutionarily conserved micro-RNA groups, with mature sequences that are identical in both humans and mice [24].

This high degree of sequence conservation between species suggests that the biological functions of miR-100-5p are also conserved, making it an ideal candidate for inter-species therapeutic applications. In the case of psoriasis,

which shares some pathological mechanisms between species, MSC-exosomes from human placenta can react with the mouse's cellular machinery. This allows the therapeutic effects observed in mice to be replicated in humans. Thus, the inter-species application of human placenta MSC-exosomes and miR-100-5p underscores the therapeutic potential of these biologics not only in animal models but also in humans. These findings lay the groundwork for translating preclinical data into clinical therapies, offering promising avenues for treating inflammatory diseases like psoriasis.

To identify potential target genes of miR-100-5p in psoriasis, we used bioinformatic predictions and literature review, identifying mTOR as a prime candidate. Studies show that the miR-99 family regulates mTOR signaling in squamous cell carcinoma and wound healing [25–27]. mTOR, part of the PI3 K/Akt pathway, influences skin cell growth, proliferation, differentiation, and motility. This pathway has gained attention as a key regulator in epidermal homeostasis and inflammatory skin disorders [28]. Akt activation is seen in psoriatic lesions, and PI3 K/Akt/mTOR also controls immune responses linked to psoriasis [19, 20, 29]. mTOR and S6 K1 activation in psoriatic skin indicates their involvement in disease pathology. Additionally, miR-100-5p inhibits angiogenesis by repressing mTOR in endothelial cells. Our results showed reduced mTOR and S6 K1 protein levels

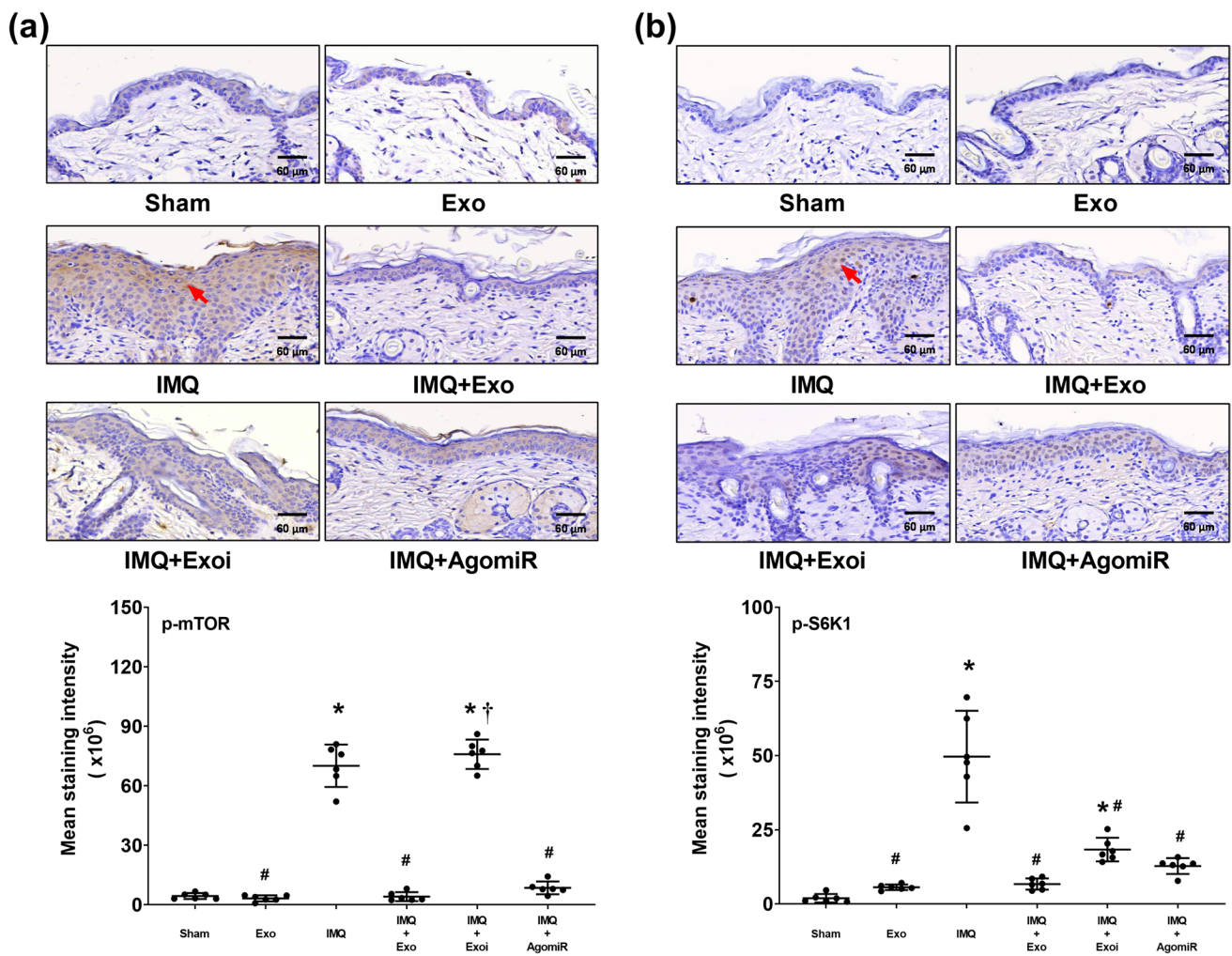


Fig. 5 The role of miR-100-5p in mediating the therapeutic effects of exosomes from human placenta mesenchymal stem cells (MSC-exosomes) in reducing imiquimod (IMQ)-induced upregulation of mammalian target of rapamycin (mTOR)/S6 kinase 1 (S6 K1) pathway in mouse skin tissues. (a) Representative microscopic images of phosphorylated-mTOR (p-mTOR, indicated by red arrows), along with the quantified total intensities of p-mTOR in mouse skin tissues, assessed via immunohistochemistry, are presented. (b) Representative microscopic images of phosphorylated-S6 K1 (p-S6 K1, indicated by red arrows), along with the quantified total intensities of p-S6 K1

in mouse skin tissues, assessed via immunohistochemistry, are presented. Tissue samples were collected the day after completing seven consecutive days of daily intervention. Data from six mice per group are presented as mean $\pm$ standard deviation. * $p < 0.05$ vs. Sham group; # $p < 0.05$ vs. IMQ group; † $p < 0.05$ vs. IMQ + Exo group. Sham: Sham group; Exo: MSC-exosome group; IMQ: imiquimod group; IMQ + Exo: IMQ plus MSC-exosomes group; IMQ + Exoi: IMQ plus MSC-exosomes treated with a miR-100-5p inhibitor; IMQ + AgomiR: IMQ plus agomiR-100-5p

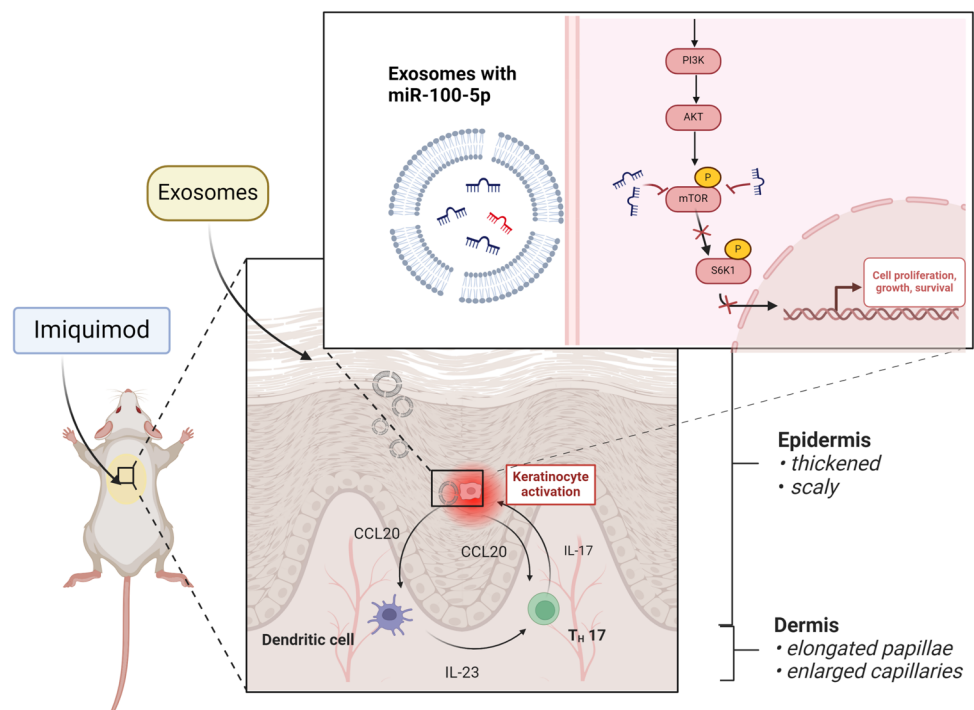
after treatment with MSC-exosomes and agomiR-100-5p, supporting the role of miR-100-5p in mitigating psoriasis induced by IMQ through targeting the mTOR/S6 K1 pathway as depicted in Fig. 6.

This study has several limitations. First, intradermal injection of agomiR-100-5p showed slightly lower therapeutic efficacy than topical MSC-exosomes, possibly due to that MSC-exosomes have better cell membrane permeate efficacy than naked microRNA, as MSC-exosomes can effectively cross the stratum corneum and reach deeper skin layers [30]. Other microRNAs in MSC-exosomes may also contribute to their therapeutic effect. Second,

the study used only male mice, so the effects in female mice with psoriasis-like dermatitis remain unknown. Third, while exosomes are effective drug delivery systems [31, 32], their content is typically low, potentially limiting efficacy. Engineering exosomes to increase miR-100-5p content could enhance their therapeutic potential for psoriasis. Further studies are needed before definitive insights can be drawn.

In conclusion, miRNA-100-5p may contribute to the therapeutic effects of human placenta MSC-exosomes against IMQ-induced psoriasis in mice by modulating the mTOR/S6 K1 pathway.

Fig. 6 Schematic diagram of the exosomes from human placenta MSC mitigating IMQ induced psoriasisiform dermatitis. Topically applying IMQ induced epidermal hyperplasia and increased cytokines in the skin tissues. Topically treated exosomes from human placenta MSC penetrated the stratum corneum to the activated keratinocytes. Exosomes from human placenta MSC which contained miR-100-5p might target the p-mTOR and the reduced the level of downstream pathway, S6 K1, caused reduced epidermal proliferation and cell growth. IMQ, imiquimod, MSC, mesenchymal stem cell, p-mTOR, phosphorylated—mammalian target of rapamycin, p-S6 K1, phosphorylated -S6 kinase 1



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Authors contribution H.Y.C.: Conceptualization, Investigation, Data curation, Writing- Original draft preparation; Funding Acquisition; C. C. Y.: Investigation, Data curation, Reviewing; H.C.J.: Supervision, Writing- Reviewing and Editing. The paper has been read and approved by all named authors.

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Data availability The data underlying this article will be shared upon reasonable request to the corresponding author.

Declarations

Institutional review board statement The Institutional Animal Use and Care Committee, Taipei Medical University, approved all the animal experiments of this study (LAC-2021–0500). The human placenta mesenchymal stem cells were a kind gift from Professor Yen-Hua Huang, Taipei Medical University, Taipei, Taiwan. Professor Yen-Hua-Huang received the ethical approval (N202101014) from the Joint Institutional Review Board of Taipei Medical University, Taipei, Taiwan, for her previous study involving placenta obtainment and isolation of human placenta mesenchymal stem cells.

Competing interests The authors have no relevant financial or non-financial interests to disclose.

References

- Menter A, Gottlieb A, Voorhees FSR, ASV, Leonardi CL, Gordon KB et al (2008) Guidelines of care for the management of psoriasis and psoriatic arthritis: section 1. J Am Acad Derm 58:826–850
- Gurunathan S, Kang MH, Jeyaraj M, Qasim M, Kim JH (2019) Review of the isolation, characterization, biological function, and multifarious therapeutic approaches of exosomes. Cell 3:8307
- Song N, Scholtmeijer M, Shah K (2020) Mesenchymal stem cell immunomodulation: mechanisms and therapeutic potential. Trends Pharmacol Sci 41:653–664
- Shin KO, Ha DH, Kim JO, Crumrine DA, Meyer JM, Wakefield JS et al (2020) Exosomes from human adipose tissue-derived mesenchymal stem cells promote epidermal barrier repair by inducing de novo synthesis of ceramides in atopic dermatitis. Cells 9:680
- Zhang B, Lai RC, Sim WK, Choo ABH, Lane EB, Lim SK (2021) Topical application of mesenchymal stem cell exosomes alleviates the imiquimod induced psoriasis-like inflammation. Int J Mol Sci 22:720
- Zhang Y, Yan J, Zheng LZJ, Sun Q (2022) Exosomes derived from human umbilical cord mesenchymal stem cells alleviate psoriasis-like skin inflammation. J Interferon Cytokine Res 42:8–18
- Zhang W, Lin J, Shi P, Su D, Cheng X, Yi W (2022) Small extracellular vesicles derived from MSCs have immunomodulatory effects to enhance delivery of ASO-210 for psoriasis treatment. Front Cell Dev Biol 10:842813
- Zhang J, Li S, Li L, Li M, Guo C, Yao J et al (2015) Exosome and exosomal microRNA: trafficking, sorting, and function. Genom Proteom Bioinform 13:17–24

9. O'Brien J, Hayder H, Zayed Y, Peng C (2018) Overview of micro-RNA biogenesis, mechanisms of actions, and circulation. *Front Endocrinol* 9:402
10. Yang X, Wang H (2021) miRNAs flowing up and down: the concerto of psoriasis. *Front Med* 8:646796
11. Su LJ, Wu MS, Hui Y, Chang YBM, Pan L, Hsu PC et al (2017) Fluorescent nanodiamonds enable quantitative tracking of human mesenchymal stem cells in miniature pigs. *Sci Rep* 7:45607
12. Lobb RJ, Becker M, Wen SW, Wong CSF, Wiegman AP, Leimgruber A et al (2015) Optimized exosome isolation protocol for cell culture supernatant and human plasma. *J Extracell Vesicles* 4:27031
13. Chen J, Chen J, Cheng YH, Fu YF, Zhao HZ, Tang MY et al (2020) Mesenchymal stem cell-derived exosomes protect beta cells against hypoxia-induced apoptosis via miR-21 by alleviating ER stress and inhibiting p38 MAPK phosphorylation. *Stem Cell Res Ther* 11:97
14. Grasso L, Wyss R, Weidenauer L, Thampi A, Demurtas D, Prudent M et al (2015) Molecular screening of cancer-derived exosomes by surface plasmon resonance spectroscopy. *Anal Bioanal Chem* 407:5425–5432
15. Chang CY, Hsu HJ, Foo J, Shih HJ, Huang CJ (2020) Peptide-Based TNF- α -binding decoy therapy mitigates lipopolysaccharide induced liver injury in mice. *Pharmaceuticals* 13:280
16. Chang CY, Chen KY, Shih HJ, Chiang M, Huang IT, Huang YH et al (2022) Let-7i-5p mediates the therapeutic effects of exosomes from human placenta chorionic membrane-derived mesenchymal stem cells on mitigating endotoxin-induced mortality and liver injury in high-fat diet-induced obese mice. *Pharmaceuticals* 15:36
17. Pantaleão LC, Ozanne SE (2018) Small RNA sequencing: a technique for miRNA profiling. *Methods Mol Biol* 735:321–330
18. Scholzen T, Gerdes J (2000) The Ki-67 protein: from the known and the unknown. *J Cell Physiol* 182:311–322
19. Calautti E, Li J, Saoncella S, Goetinck PF (2005) Phosphoinositide 3-kinase signaling to Akt promotes keratinocyte differentiation versus death. *J Biol Chem* 280:32856–32865
20. Huang T, Lin X, Meng X, Lin M (2014) Phosphoinositide-3 kinase/protein kinase-B/mammalian target of rapamycin pathway in psoriasis pathogenesis. A potential therapeutic target? *Acta Derm Venereol* 94:371–379
21. Ivey KN, Srivastava D (2015) microRNAs as developmental regulators. *Cold Spring Harb Perspect Biol* 7:a008144
22. Jiang X, Shi R, Ma R, Tang X, Gong Y, Yu Z et al (2023) The role of microRNA in psoriasis: a review. *Exp Dermatol* 32:1598–1612
23. Joyce CE, Zhou X, Xia J, Ryan C, Thrash B, Menter A et al (2011) Deep sequencing of small RNAs from human skin reveals major alterations in the psoriasis miRNAome. *Hum Mol Genet* 20:4025–4040
24. Hertel J, Bartschat S, Wintsche A, Otto C (2012) Evolution of the let-7 microRNA Family. *RNA Biol* 9:231–241
25. Chen Z, Jin Y, Yu D, Wang A, Mahjabeen I, Wang C et al (2012) Down-regulation of the microRNA-99 family members in head and neck squamous cell carcinoma. *Oral Oncol* 48:686–691
26. Sun J, Chen Z, Tan X, Zhou F, Tan F, Gao Y et al (2013) MicroRNA-99a/100 promotes apoptosis by targeting mTOR in human esophageal squamous cell carcinoma. *Med Oncol* 30:411
27. Jin Y, Tymen SD, Chen D, Fang ZJ, Zhao Y, Dragas D et al (2013) MicroRNA-99 family targets AKT/mTOR signaling pathway in dermal wound healing. *PLoS ONE* 8:e64434
28. Buerger C, Shirsath N, Lang V, Berard A, Diehl S, Kaufmann R et al (2017) Inflammation dependent MTORC1 signaling interferes with the switch from keratinocyte proliferation to differentiation. *PLoS On* 12:e0180853
29. Buerger C, Malisiewicz B, Eiser A, Hardt K, Boehncke WH (2013) Mammalian target of rapamycin and its downstream signaling components are activated in psoriatic skin. *Br J Dermatol* 169:156–159
30. Ma H, Siu WS, Leung PC (2023) The potential of MSC-based cell-free therapy in wound healing-a thorough literature review. *Int J Mol Sci* 24:9356
31. Kim G, Kim M, Lee Y, Byun JW, Hwang DW, Lee M (2020) Systemic delivery of microRNA-21 antisense oligonucleotides to the brain using T7- peptide decorated exosomes. *J Control Release* 317:273–281
32. Yang J, Luo S, Zhang J, Yu T, Fu Z, Zheng Y et al (2021) Exosome mediated delivery of antisense oligonucleotides targeting α -synuclein ameliorates the pathology in a mouse model of parkinson's disease. *Neurobiol Dis* 148:105218

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